HAMDI CAN ERNALBANTOĞULLARI

SOFTWARE TEST ENGINEER



QA & Test Automation | Python · Pytest · Playwright · Selenium | API Testing | System Analysis | AI & LLM-Assisted Testing

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SUMMARY

Software Test Engineer with 3+ years of professional software testing and quality assurance experience in consumer electronics R&D, supported by an engineering background in system design, embedded software, and hardware-software integrated products. Experienced across the Software Testing Life Cycle (STLC), including functional, regression, smoke/sanity and exploratory testing, test design, defect lifecycle management, release validation, production/field issue investigation, and real-device Smart TV testing at Vestel R&D.

Hands-on Test Automation and API Testing experience through technical projects using Python, Pytest, Playwright, Selenium WebDriver, REST APIs, Postman/Newman, POM, Git, GitHub, and GitHub Actions. Experience with Generative AI and LLM-assisted engineering workflows includes Codex and Claude Code for repository analysis, test ideation, debugging, code understanding, documentation, and rapid prototyping.

Core competencies: Software Testing & QA · Test Automation · API Testing · Defect Lifecycle Management · Test Analysis · System Analysis · Production & Field Issue Investigation · Release Validation · Real-Device Testing

WORK EXPERIENCE

Software Test Engineer, Vestel Nov 2022 – Nov 2025

Manisa, Türkiye | Hybrid

- Validated Smart TV software across 9 main projects and 20+ panel/RAM variant sub-projects, including 3 greenfield projects taken into production, 5 maintenance projects, and 1 finalised project, through functional, regression, smoke/sanity, exploratory, build, and release validation testing.
- Planned an average of 2 release test cycles per week, performing release candidate validation and targeted regression; identified defects causing release holds and new build requests, and verified fixes before release continuation.
- Designed and executed tens of test cases for maintenance projects and thousands for greenfield R&D projects based on functional/technical requirements, edge cases, negative scenarios, and failure conditions; analysed behaviour to determine test conditions, quality risks, and validation approaches.
- Handled 4,000+ defect records across internal testing, production QC, and automated testing using Jira, covering reproduction, investigation, fix verification, regression testing, and closure throughout the defect lifecycle.
- Prepared defect reports with reproduction steps, expected/actual results, technical evidence, severity/priority, affected builds, reproduction conditions, and release impact; analysed diagnostics, logs, system behaviour, and software/build versions for root cause investigation and system analysis.
- Investigated 150+ mass-production and field issues, primarily involving production stops and customer complaints, reproducing issues in controlled environments and validating fixes.
- Collaborated across Software, Hardware, Quality Control, Production, and Customer Support, communicating defects, quality risks, technical findings, and release impact through investigation, verification, and closure.

System Design Engineer, Vestel Jul 2021 – Nov 2022

Manisa, Türkiye

- Designed electronic systems for LED display modules and cabinet structures, analysing technical, system-level, hardware, and product requirements.
- Supported product development, system/design verification, and technical validation, alongside technical problem solving, documentation, project coordination, system/requirements analysis, and hardware-software interaction.

INTERNSHIP EXPERIENCE

Software Engineer Intern, DVLX R&D Engineering Company Jul 2019 – Aug 2019

- Participated in embedded software development, module-level software testing and verification, debugging, and technical problem solving, with exposure to embedded systems, SDLC, and hardware-software interaction.

Software Engineer Intern, Esinotek R&D Engineering Inc. Jul 2018

- Participated in R&D activities involving technical research, engineering analysis, audio/signal processing, electrochemical analysis, and technical problem solving.

TECHNICAL SKILLS

Software Testing & QA: Functional Testing · Regression Testing · Smoke/Sanity Testing · Exploratory Testing · Design Verification · Real-Device Testing · Android-Based Platform Testing · Test Case Design · Test Scenario Design · Build Verification · Release Validation · Fix Verification · STLC

Test Automation: Python · Pytest · Playwright · Selenium WebDriver · POM · Pytest Fixtures · UI Automation · Smoke & Regression Suites · JUnit XML Reporting · Playwright Trace Viewer

API Testing: REST APIs · Python Requests · Postman · Newman · JSON · API Client Layers · Request/Response Validation

Defect & Quality Management: Jira · Defect Management · Defect Reporting · Defect Tracking · Defect Lifecycle Management · Severity/Priority Assessment · Log Analysis · Quality Risk Analysis · Release Impact Assessment

Analysis & Engineering: Requirements Analysis · System Analysis · Test Analysis · Root Cause Investigation · Hardware-Software Interaction · Technical Problem Solving · System Design · Product Development · Technical Documentation · Project Coordination · Stakeholder Communication · Cross-Functional Collaboration · Release Coordination · MATLAB · Simulink

AI & LLM-Assisted Engineering: Generative AI · LLMs · AI-Assisted Testing · Repository Analysis · Test Idea Generation · Test Coverage Exploration · AI-Assisted Debugging · Codex · Claude Code · GPT · Gemini

CI/CD & Development Tools: Git · GitHub · GitHub Actions · Chrome DevTools · CI/CD · Software Development Life Cycle (SDLC)

TECHNICAL PROJECTS

Python Test Automation Portfolio

Python · Pytest · Playwright · Selenium WebDriver · REST APIs · Postman · Newman · GitHub Actions

- Automated 15 documented Playwright UI scenarios across TodoMVC and SauceDemo using Python, Pytest, POM, reusable fixtures, and smoke/regression markers, with SauceDemo flows also implemented through Selenium WebDriver.
- Developed API automation using Python Requests, Pytest, and REST APIs, covering request/response validation, JSON handling, reusable API client layers, and Postman/Newman execution.
- Configured 3 GitHub Actions workflows for Pytest UI/API automation, Newman execution, and CodeQL analysis, with JUnit XML, screenshots, logs, Playwright traces, and failure artifacts supporting CI-based investigation.

WasteWisely — Digital Waste Auditor

Python · FastAPI · Pytest · Playwright · pywebview · Generative AI

- Developed a Windows desktop application using Python and FastAPI for directory scanning, waste classification, sensitive-file detection, cleanup, and archiving; implemented Pytest/Playwright testing, manual/workflow/packaging validation, and Codex-assisted repository analysis and debugging with human verification.

OKX Trading Bot

Python · Pytest · API Integration

- Developed a modular Python spot-trading application with technical scoring, market-regime detection, risk controls, reconciliation, Take Profit / Stop Loss, trade sizing, and Telegram monitoring, supported by Pytest-based automated tests covering execution, reconciliation, risk controls, market-regime behaviour, and trade sizing.

ACADEMIC PROJECTS

ECG Arrhythmia Classification — MATLAB — ANN-based ECG arrhythmia classification for biomedical signal analysis.

Traffic Sign Recognition — MATLAB — Neural-network-based traffic sign recognition and classification.

Speech Signal Processing — MATLAB · STFT — STFT-based speech analysis and voiced/unvoiced segment classification.

AI & LLM-ASSISTED ENGINEERING

AI-Assisted QA: Test idea/scenario generation · Test coverage exploration · Failure investigation · Debugging

Engineering Workflows: Repository/codebase analysis · Code understanding · Technical investigation · Documentation · Rapid prototyping

Tools: Codex · Claude Code · GPT · Gemini

EDUCATION

Izmir Institute of Technology Sep 2015 – Mar 2021
Bachelor of Science — Electrical and Electronics Engineering
Electronics and Communication Program

CERTIFICATIONS & TRAINING

Artificial Intelligence Specialization Training — Opsiyon Akademi / Riscon Academy
Skills: Artificial Intelligence · Python

ISTQB 2018 Foundation Level (CTFL) Exam Preparation — Udemy
Skills: Software Testing · Manual Testing · Test Management

Generative AI — GPT, Gemini, Claude & MCP — Udemy
Skills: Generative AI · Large Language Models · GPT · Gemini · Claude · MCP · AI-Assisted Engineering

LANGUAGES

Turkish: Native
English: Professional Working Proficiency C1
Italian: CILS A1

PROFESSIONAL INTERESTS

Software Test Engineering · Quality Assurance · Test Automation · API Testing · Test Analysis · System Analysis · Business Analysis · Technical Project Coordination · AI-Assisted QA

REFERENCES

Available upon request.